

CELLCHECK · DESIGN DECISION RECORD

CellCheck TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	CCK-DDR-001	Version	0.3
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/cellcheck		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	Items 10, 12, 14 and 16 decided by Amish on 2026-10-02 as recommended in CCK-DEC-001
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review items adopted for TRL 3 under Amish's 2026-09-25 instruction, open for his review, and the items that stay open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** decided for items with a recommendation. On 2026-09-25 Amish accepted all recommendations (CCK-DDR-002): items 1 to 9, 11, 13 and 15 are Decided by Amish, 2026-09-25: go with recommendation. Items 10, 12, 14 and 16, which had no recommendation here, were given recommendations in the design decisions register (CCK-DEC-001, open items 3 to 6) and accepted by Amish on 2026-10-02: "i approve your recommendations for all 555 open decisions."

■ Context

The TRL 2 review note (docs/REVIEW.md , session of 2026-09-25) listed ten items as "Proposed, awaiting Amish", nine of them with a recommendation. The TRL 3 session first adopted each recommended item for TRL 3 work, open for Amish's review, and added items 11 to 16. Later on 2026-09-25 Amish wrote "i accept all your recommendations, go with them across all repos." Each item that carries a recommendation is therefore decided as recommended; where several options were offered, the recommended option is the decision. What changed in the repo is listed in CCK-DDR-002. TRL 4 is on hold by Amish's instruction.

■ Options considered

The options for each item are in docs/REVIEW.md (TRL 2 session) and CCK-PRC-001 v0.2. They are not repeated here.

■ Decision

Table 1. Decided items.

#	Item	Status and content	Where it now lives
1	SwapCell link (affects the pitch)	Decided by Amish, 2026-09-25: go with recommendation. Option (a): keep the pitch and plan a second-life SwapCell variant, agreed with the SwapCell project. The TRL 3 study (CCK-CAL-001 section 10) finds a low-current 13S4P storage pack for PowerBox feasible at about 334 Wh and 165 W; e-bike use is ruled out. No pitch rewording was recommended, so <code>project.yaml</code> and the README keep their pitch and problem lines. The agreement with SwapCell is a cross-repo action	CCK-PRC-001, CCK-PRB-001, CCK-CAL-001 section 10, README
2	Unattended operation (safety trade-off)	Decided by Amish, 2026-09-25: go with recommendation. Charge and discharge are attended only; the rest stage (no current flowing) may be left unattended	CCK-REQ-001 R14 (redefined), CCK-PRC-001 safety, README
3	Single-fault protection	Decided by Amish, 2026-09-25: go with recommendation. A hardware watchdog removes all charge enables and load gate drive when the controller stops, and a per-channel undervoltage comparator removes load gate drive below 2.5 V, about \$5	<code>bom/bom.csv</code> line 18, <code>cad/src/model.py</code> part 18, CCK-CAL-001 section 6, CCK-REQ-001 R9
4	Budget	Decided by Amish, 2026-09-25: go with recommendation. Option (a): <code>budget_usd</code> raised from \$160 to \$175 for the watchdog and margin. The \$164 BOM now meets R12 with \$11 of margin	<code>project.yaml</code> , CCK-REQ-001 v0.4 R12, CCK-CAL-001 v0.2 section 11
5	Grading thresholds, 14-day rest with a 50 mV limit, $\pm 1\%$ group tolerance	Decided by Amish, 2026-09-25: go with recommendation. Starting values, to be tuned with the first 100 real cells; the tuning is TRL 4 work and on hold	CCK-PRC-001 Table 2, CCK-REQ-001 R6 and R15
6	Module choices	Decided by Amish, 2026-09-25: go with recommendation. TP5100-class charger, INA226-class monitor with a 25 m Ω shunt, ESP32-S3 class controller	CCK-PRC-001 Table 1, <code>bom/bom.csv</code> lines 6, 7 and 11
7	Channel count and load type	Decided by Amish, 2026-09-25: go with recommendation. Eight channels with a linear load; 16 channels and regenerative discharge are later variants	CCK-PRC-001
8	LFP profile	Decided by Amish, 2026-09-25: go with recommendation. Later, after the lithium-ion profile is proven	CCK-PRC-001, CCK-PRB-001
9	Record format	Decided by Amish, 2026-09-25: go with recommendation. The per-cell CSV stays the primary record; each matched group and each reject lot maps to ReflowEconomy's material passport v0.2 (two schema gaps, see item 15)	CCK-CAL-001 section 13, CCK-PRC-001
11	DC four-wire resistance rather than 1 kHz AC impedance, and the 14-day (rather than 7-day) rest	Decided by Amish, 2026-09-25: go with recommendation. DC pulses and a 14-day rest, as recommended in the precis and used as the working basis of CCK-CAL-001	CCK-PRC-001, CCK-CAL-001

#	Item	Status and content	Where it now lives
13	TRL 3 engineering proposals from CCK-CAL-001	Decided by Amish, 2026-09-25: go with recommendation. Firmware and build rules: read self-discharge in the same channel and against the median of the same-day cohort (R6); start the resistance pulse only within 1 K of the bench (R4, now met at ± 2.09 m Ω); per-channel two-point voltage calibration at build (R3); stop discharges on a fan or heatsink fault (R8); a plausibility check for a detached thermistor (R9). Writing and testing the firmware and doing the calibration are TRL 4 work and on hold	CCK-PRC-001 v0.4, CCK-CAL-001 v0.2 section 5, CCK-REQ-001 v0.4
15	ReflowEconomy passport gaps: no <code>product_form</code> for graded cells and no electrical test in <code>identification.method</code>	Decided by Amish, 2026-09-25: go with recommendation. Raise with the ReflowEconomy project; listed under cross-repo actions in <code>docs/REVIEW.md</code> ; this repo does not edit that schema	CCK-CAL-001 section 13, <code>docs/REVIEW.md</code>

Items decided on 2026-10-02

Table 2. Items that had no recommendation here; decided by Amish on 2026-10-02 as recommended in the design decisions register (CCK-DEC-001).

#	Item	Status
10	First trial partner: a repair café, an e-bike repair shop or a battery collection point	Decided 2026-10-02: a local Repair Café group that already passes laptop batteries to an e-waste or battery collection point, with that collection point as the source of trial cells; the first candidate to approach, not yet agreed (CCK-DEC-001, item 3)
12	Scheduling rule for the attended day: current stages start only if they finish the same day (12.0 cells per day), or pause overnight with the unit switched off (12.8)	Decided 2026-10-02: a charge or discharge starts only if it will finish the same day (12.0 cells per day); the overnight pause is not used, as it adds an unquantified capacity error (CCK-DEC-001, item 4)
14	Responses to the requirements at risk: R11 (mass; a 1.2 mm tray saves about 0.49 kg but thins the containment) and R5 (throughput). R12 is now met through item 4	Decided 2026-10-02: R11, accept 4.91 kg, keep the 1.5 mm tray and weigh the prototype at TRL 4; R5, the same-day rule of item 12, with more channels left for a later version (CCK-DEC-001, item 5)
16	Grade A resistance limit per cell model (60 m Ω default)	Decided 2026-10-02: keep 60 m Ω as the default; add per-model limits only for the few cell models most common in the partner's intake, set from DC resistance measured on known-good cells of each model, not from data sheet impedance (normally 1 kHz AC values) (CCK-DEC-001, item 6)

Consequences

- R9 is met on analysis, and R14 is met by design as redefined. The original R14 target (overnight grading without a person present) is superseded, not met.
- The budget is \$175, so R12 is met at \$164.
- The temperature-gated resistance pulse makes R4 met on the error budget (± 2.09 m Ω against ± 3.0 m Ω).

- Attended-only operation makes throughput depend on scheduling; R5 is at risk at exactly 12.0 cells per day under the same-day scheduling rule decided on 2026-10-02 (item 12).
- The second-life SwapCell variant needs the SwapCell project's agreement and a pack design of its own. CellCheck supplies graded cells and records only.
- TRL 4 is on hold by Amish's instruction.